Ian Yi Shang

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Education

University of Toronto — Toronto, Ontario, Canada

BASc Mechanical Engineering | Professional Experience Year Co-op — *Sep. 2024 – Jun. 2029*

Programs: Environmental Engineering Minor, Electric Vehicle Design Certificate, Forensics Engineering Certificate

Courses: Manufacturing, Forensic Engineering, Statistics, Kinematics and Dynamics of Machines, Solid Mechanics, Dynamics

GPA: 3.39

Experience

MGT100 Teaching Assistant — Dec. 2024 – Present

University of Toronto, Rotman School of Management — *Toronto, Ontario, Canada*

- Coordinate logistics by preparing rooms, checking in students, and collecting materials, supporting efficient exams.
- Oversee exam procedures, conduct student sign-ins, and answer student questions during exams, maintaining organized testing environment.

Mechanical Engineer Intern — May 2026 – Jul. 2026

Fudan University Robotics Research Association — *Shanghai, China*

- Teach myself SolidWorks to model and assemble a custom 200x220mm robotic vehicle chassis and servo mounts, integrating wheels, actuators, battery, and control electronics into a functional prototype.
- Used SolidWorks to design and 3D-print a Python-based, IMU-guided two-servo self-righting mechanism featuring closed-loop orientation control, adaptive motion profiling, stall protection, and automatic serial reconnection.

SHAD Member — Jul. 2023

SHAD Ontario Tech University — *Oshawa, Ontario, Canada*

- Developed and prototyped cost-efficient housing solutions for homeless populations across Canada.
- Designed a tower for an earthquake-resistance challenge, earning first place in the structural design competition.
- Built and 3D-printed wind turbine blades in a power generation efficiency competition, placing second.
- Built a computer vision model to automatically detect whether individuals were wearing protective face masks.

Projects

Minecraft Biome Recognition CNN | *Python, PyTorch, AI, DL, Data Analysis* — May 2026 – Aug. 2026

- Developed a PyTorch CNN to classify self-collected Minecraft screenshots into eight biome classes using data augmentation, weight decay, and train/validation/test splits to improve generalization.
- Benchmarked and analyzed the CNN against color histogram, RGB, ANN, and ResNet models, achieving 84.55% test accuracy.

Beginner Cinematographer Robotic Arm | *CAD, SolidWorks, Design* — Sep. 2025 – Dec. 2025

- Designed a 1.2 m-reach, six-axis robotic arm in SolidWorks, integrating transmission components, motors, bearings, aluminum structures, and 3D-printed parts within a sub-$1,000 CAD estimate.
- Performed load-based torque calculations and SolidWorks Motion Analysis for a six-axis robotic arm, evaluating performance under a conservative 12 kg simulated end load.
- Compared five mechanical concepts using a Pugh matrix; refined materials, structural layout, and transmission choices for weight, cost, and manufacturability.

Landmark Garage Advertising Campaign | *Data Analysis, Marketing* — Jan. 2025 – Apr. 2025

- Developed project requirements, conceptual design specifications, and a team charter while collaborating with clients in multiple meetings to define problem statements and find solutions around the clients' needs.
- Design an Instagram campaign page with 27,000 views for the University of Toronto Landmark Garage to increase its usage and present the final solution to clients, TAs, and other students within a 12 minute time constraint.

Skills, Awards, Hobbies

Industry Tools: Python, MS Office
Hands-on Skills: Machining experience from George Brown Polytechnic, experience with power tools and hand tools
Awards: Dean's List, Chem 13 News Exam Outstanding Achievement, AP Scholar with Distinction Award, Honor Roll Award (Grade 10, 11), Principal's Honor Roll (Grade 12), Geography 12 Award
Interpersonal Skills: Presentation, Public Speaking, Communication, Teamwork, Collaboration, Conflict Resolution
Certificates: Swim Instructor, Bronze Medallion & Cross, Red Cross Babysitting
Hobbies: Photography, assembling and painting military models, building technical and architectural Minecraft